Blackpill vs Bluepill:
1.Blackpill internal:
1a.FETTEMP ADC channel change: isolate RST trace from pin 5, solder pin 5 to pin 13
2.Blackpill interface:
2a.Solder UART1 connectors to pin 34 & 35
2b.UART3 not available, can be used as GPIO B1 & B2
3.Unwanted capacitor(C6,100n) on PB10, probably harmless
4.All other interfaces identical
5.All earlier used ADC channels (except FETTEMP) numbers at the processor to be subtracted by one
6.Blackpill internal change: remove diode inline with 5V & short the pads

U1
Blackpill
+5V
+3V3
GND
100n C6
PB12 PB13 PB14 PB15 PA8 PA9 PA10 PA11 USB− PA12 USB+ PA15 PB3 PB4 PB5 PB6 TX1 PB7 RX1 PB8 PB9
PB10 PB2 PB1 PB0 PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0 RST PC15 PC14 PC13 LED
PWMALOW PWMBLOW PWMCLOW PWMAHIGH PWMBHIGH PWMCHIGH
B2 B1 FETTEMP ADC12_IN7 ADC12_IN6 PHASECCURRENT PHASEBCURRENT PHASEACURRENT BATTERYVOLTAGE MOTORTEMP HALLSENSORC HALLSENSORB HALLSENSORA
USART1TX USART1RX
sink 3mA source 0mA 30pF 2MHz
PWM pin (F4xx compatible)
5V tolerant

U3
STH0548S12
VBAT Vin Vout +12V
GND GND GND
DISABLEPOWER REM TRIM
C8 10u

U4
MP1584EN_4pin
IN+ IN− OUT+ OUT− +5V
C15 10u C12 10u C14 10u

VBAT R1 1M BATTERYVOLTAGE R2 33K GND

+3V3 R25 10K FETTEMP
10K NTC1 C10 100n GND

+3V3 R5 2.2K J8 ADC12_IN7 ADC12_IN6 GND

+5V +3V3
R31 2.2K R32 2.2K R33 2.2K R4 10K MOTORTEMP
J5 HALLSENSORC HALLSENSORB HALLSENSORA
R6 100 R8 100 R9 100
C19 100n C20 100n C21 100n C22 100n GND

J7 B1 B2 GND
3V3 5V +3V3 +5V

VBAT XC0 680u XC1 100u XC2 100u XC3 100u XC4 2.2u XC5 2.2u XC6 2.2u XC7 2.2u XC8 2.2u XC9 2.2u GND

+3V3 C7 10u C13 10u C16 10u C18 10u GND

+12V C4 10u GND
PWMAHIGH PWMALOW
U5 U3216 VCC HIN LIN COM VB HO VS LO
SM4007PL D1
BOOT_A
10u C1 DRVH_A DRVL_A
4.7 RG1 GH_A Q1 MDP10N027 10n OC1
4.7 RG2 GL_A Q2 MDP10N027 10n OC2
2.2 RS1 1n CS1
SH_A 0.001 SHA GND
VBAT
PHASEA J1 Conn_01x01
2.2K R20 R19 2.2K
R12 22K U2A NCS20034DR2G R10 100 PHASEACURRENT 22K R7

+12V C9 10u GND
PWMBHIGH PWMBLOW
U6 U3216 VCC HIN LIN COM VB HO VS LO
SM4007PL D2
BOOT_B
10u C2 DRVH_B DRVL_B
4.7 RG3 GH_B Q3 MDP10N027 10n OC3
4.7 RG4 GL_B Q4 MDP10N027 10n OC4
2.2 RS2 1n CS2
SH_B 0.001 SHB GND
VBAT
PHASEB J2 Conn_01x01
2.2K R22 R21 2.2K
R15 22K U2D NCS20034DR2G R11 100 PHASEBCURRENT 22K R16

+12V C11 10u GND
PWMCHIGH PWMCLOW
U7 U3216 VCC HIN LIN COM VB HO VS LO
SM4007PL D3
BOOT_C
10u C3 DRVH_C DRVL_C
4.7 RG5 GH_C Q5 MDP10N027 10n OC5
4.7 RG6 GL_C Q6 MDP10N027 10n OC6
2.2 RS3 1n CS3
SH_C 0.001 SHC GND
VBAT
PHASEC J3 Conn_01x01
2.2K R23 R24 2.2K
R17 22K U2C NCS20034DR2G R13 100 PHASECCURRENT 22K R18

VBAT J9
GND J10
+12V J11
DISABLEPOWER J12
VBAT J4
GND J6
PWR_FLAG VBAT

+3V3 R3 2.2K VREF
100n C5 R14 2.2K GND
U2B NCS20034DR2G VBIAS
+3V3 U2E NCS20034DR2G V+ V− 11 GND

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Title: Easy DIY ESC
Size: A3 | Date: 2022-10-12 | Rev:
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